Analysis of arthopod communites in Hawaii using METE Theory

This supplement walks through the steps of recreating the entire analysis presented in the main text. It also provided further analysis in support of and to contextualize results in the main text.

Computing setup

Analyses are supported by a custom R package *HawaiiArthMETE* [@foo] which can be installed from github.

```
devtools::install_github("ecoevomatics/HawaiiArthMETE")
```

The quarto [@quarto] document used to generate this supplement can be found with the R command `system.file("ms/hawaii-arth-mete-supp.qmd", package = "HawaiiArthMETE")`.

The following R computing environment is used for all analyses:

```
library(meteR)
library(ggplot2)
library(ggh4x)
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(vegan)
```

```
Loading required package: permute
```

```r
library(metafor)
```

```
Loading required package: Matrix


Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack

Loading required package: metadat

Loading required package: numDeriv


Loading the 'metafor' package (version 5.0-1). For an
introduction to the package please type: help(metafor)


Attaching package: 'metafor'

The following object is masked from 'package:vegan':

    permutest
```

```r
library(HawaiiArthMETE)
```

```
Attaching package: 'HawaiiArthMETE'

The following object is masked from 'package:meteR':

    arth
```

```r
knitr::opts_chunk$set(
    # default figure dims
    fig.width = 5.5, fig.height = 4,
    # caching is default for all subsequent chunks
    cache = TRUE
)

# load data
data(arth)

sessionInfo() |>
    print(locale = FALSE)
```

```
R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.0.1

Matrix products: default
BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framewo
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] HawaiiArthMETE_0.1.0 metafor_5.0-1        numDeriv_2016.8-1.1
 [4] metadat_1.6-0        Matrix_1.7-4         vegan_2.7-2
 [7] permute_0.9-8        tidyr_1.3.1          dplyr_1.2.1
[10] ggh4x_0.3.1          ggplot2_4.0.1        meteR_1.2

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       jsonlite_2.0.0     compiler_4.5.2     tidyselect_1.2.1
 [5] parallel_4.5.2     cluster_2.1.8.1    splines_4.5.2      scales_1.4.0
 [9] yaml_2.3.10        fastmap_1.2.0      lattice_0.22-7     R6_2.6.1
[13] generics_0.1.4     knitr_1.50         MASS_7.3-65        tibble_3.3.0
```

```
[17] pillar_1.11.0      RColorBrewer_1.1-3 rlang_1.2.0        mathjaxr_2.0-0
[21] xfun_0.52          S7_0.2.1           cli_3.6.5          mgcv_1.9-3
[25] withr_3.0.2        magrittr_2.0.4     digest_0.6.39      grid_4.5.2
[29] rstudioapi_0.17.1  nlme_3.1-168       lifecycle_1.0.5    vctrs_0.7.3
[33] evaluate_1.0.4     glue_1.8.0         farver_2.1.2       rmarkdown_2.29
[37] purrr_1.1.0        tools_4.5.2        pkgconfig_2.0.3    htmltools_0.5.8.1
```

METE calculations

Comparison of the observed data to predictions from the Maximum Entropy Theory of Ecology (METE) is performed with package *meteR* [@rominger-merow2016]. The following code records information about each sample (name, flow age, proportion non-native presence) as well as several metrics of deviation between theory and observation:

- a z^2 value measuring a normalized difference in likelihoods between observed data and theoretical expectation
- differences in proportion of rarity (number of singletons or normalized metabolic rate ≤ 2) observed and expected
- differences in proportion of dominance (sum of top three most abundant species or sum of top three highest normalized metabolic rates) observed and expected

All metrics are calculated for both the species abundance distribution (SAD) and individual metabolic rate (or power) distribution (IPD).

```r
arth_mete <- split(arth, arth[, c("site", "tree")], drop = TRUE) |>
    lapply(function(x) {
        # the esf for this grouping
        esf <- meteESF(spp = x$species_code,
                       abund = x$abundance,
                       power = x$metabolic_rate)

        # sad and ipd
        s <- sad(esf)
        p <- ipd(esf)

        data.frame(
            # sample info
            site = x$site[1],
            site_name = x$site_name[1],
            tree = x$tree[1],
            flow_age_my = x$flow_age_my[1],
```

```
            # z2 values
            z2_sad = logLikZ(s)$z,
            z2_ipd = logLikZ(p)$z,

            # prop non-native for abundance and power
            prop_n_non_nat = sum(
                x$abundance[!(x$origin %in% c("endemic", "indigneous"))]
            ) /
                sum(x$abundance),
            prop_p_non_nat = sum(
                x$metabolic_rate[!(x$origin %in% c("endemic", "indigneous"))]
            ) /
                sum(x$metabolic_rate),

            # differences in extreme values
            n1_diff = log(mean(meteDist2Rank(s) == 1) / mean(s$data == 1)),
            nmax_diff = log(sum(meteDist2Rank(s)[1:10]) / sum(s$data[1:10])),
            p1_diff = log(mean(meteDist2Rank(p) <= 3) / mean(p$data <= 3)),
            pmax_diff = log(sum(meteDist2Rank(p)[1:10]) / sum(p$data[1:10]))
        )
    })

arth_mete <- do.call(rbind, arth_mete)
```

An initial investigation of patterns in the SAD reveal two clear outliers in terms of deviation from theory and proportion of non-native individuals (Figure 1).

Investigating highly invaded, highly dominated samples

The two outlier samples both show a difference (theory - observed) in maximum abundance of over -300 (Figure 1).

```
outliers <- filter(arth_mete, nmax_diff < -300)
```

```
pivot_longer(outliers,
             c(prop_n_non_nat, nmax_diff),
             names_to = "metric")
```

```
# A tibble: 4 x 12
  site   site_name  tree flow_age_my z2_sad z2_ipd prop_p_non_nat n1_diff p1_diff
```

```
  <fct> <fct>     <int>       <dbl>  <dbl>  <dbl>          <dbl>   <dbl>   <dbl>
1 LA    Laupāhoe~     2       0.005   7.95 27.3            0.395  -0.157  0.180
2 LA    Laupāhoe~     2       0.005   7.95 27.3            0.395  -0.157  0.180
3 KH    Kohala        6       0.15    7.09  0.241          0.575  -0.186 -0.0392
4 KH    Kohala        6       0.15    7.09  0.241          0.575  -0.186 -0.0392
# i 3 more variables: pmax_diff <dbl>, metric <chr>, value <dbl>
```

```
select(arth_mete, site_name, prop_n_non_nat, nmax_diff, z2_sad) |>
    # pivot so both metrics are plotted in faceted panels
    pivot_longer(prop_n_non_nat:nmax_diff, names_to = "metric") |>
    ggplot(aes(value, z2_sad, color = site_name)) +
    geom_point() +

    # add arrows pointing to outliers
    geom_segment(
        data = pivot_longer(outliers,
                            c(prop_n_non_nat, nmax_diff),
                            names_to = "metric"),
        aes(x = value * 0.8, y = z2_sad * 0.9,
            xend = value * 0.97, yend = z2_sad * 0.99),
        arrow = arrow(length = unit(1, "char")),
        inherit.aes = FALSE
    ) +

    # facet by metric and trick strip labels into being
    # axis labels
    facet_grid(
        cols = vars(metric), scales = "free_x",
        labeller = as_labeller(
            c(nmax_diff = "Diff. in maximum abundance",
              prop_n_non_nat = "Prop. non-native individuals")
        ),
        switch = "x"
    ) +
    ylab(expression("SAD"~z^2*"-value")) +
    this_ms_look() +
    theme(strip.background = element_blank(),
          strip.placement = "outside",
          axis.title.x = element_blank())
```

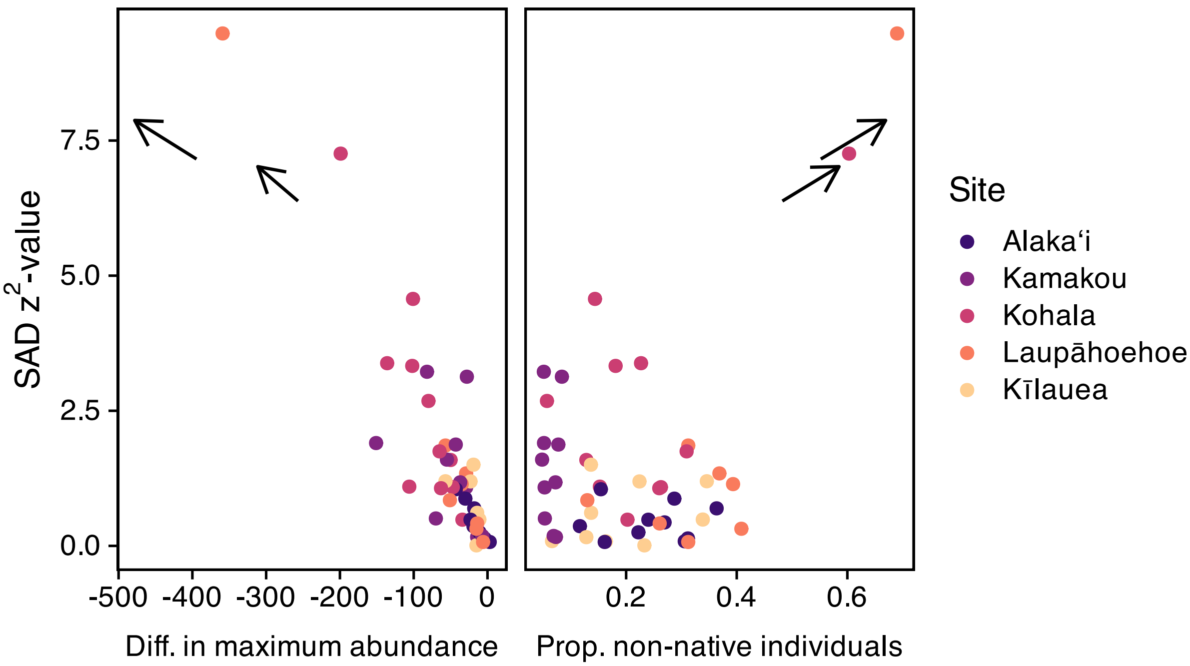

Figure 1

These two outlier samples come from the Laupāhoehoe and Kohala sites. To further understand these two samples and determine if they should remain in the analyses we look at their composition. Table 1 shows the three most abundant species for the Laupāhoehoe outlier sample while Table 2 shows the three most abundant species for the ‘r Kohala outlier sample.

```
filter(arth, tree == outliers$tree[1] & site == outliers$site[1]) |>
    summarize(n = n(), .by = c(genus, species, origin)) |>
    arrange(desc(n)) |>
    head(3) |>
    knitr::kable(
        caption = paste(
            "Top three most abundant taxa from",
            levels(arth$site_name)[as.numeric(outliers$site[1])],
            "outlier sample"
        )
    )
```

Table 1: Top three most abundant taxa from Laupāhoehoe outlier sample

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genus	species	origin	n
Greenidea	formosana	adventive	423
Salina	celebensis	adventive	330
Entomobrya	socia	adventive	69

```
filter(arth, tree == outliers$tree[2] & site == outliers$site[2]) |>
    summarize(n = n(), .by = c(genus, species, origin)) |>
    arrange(desc(n)) |>
    head(3) |>
    knitr::kable(
        caption = paste(
            "Top three most abundant taxa from",
            levels(arth$site_name)[as.numeric(outliers$site[2])],
            "outlier sample"
        )
    )
```

Table 2: Top three most abundant taxa from Kohala outlier sample

Table 2: Top three most abundant taxa from Kohala outlier sample

genus	species	origin	n
Salina	celebensis	adventive	410
Forcipomyia	hardyi	endemic	92
Kilauella	sp_3	endemic	46

In both outliers the most abundant species is non-native, and in the case of Laupāhoehoe, all top three most abundant species are non-native. Laupāhoehoe and Kohala share *Salina celebensis* in common which is a Collembola known to reach high densities in Hawaiʻi [@christiansen1992]. The abundances of *Salina celebensis* and *Greenidea formosana* in these samples are significant outliers compared to the maximum abundance found in any other sample as shown in Figure 2.

```
crazy_abund <- filter(arth,
                      (site %in% outliers$site & tree %in% outliers$tree)) |>
    summarize(n = n(), .by = c(site, tree, species_code)) |>
```

```
    arrange(desc(n)) |>
    head(3)

summarize(arth, .by = c(site, tree),
          nmax = max(table(species_code))) |>
    filter(nmax < 300) |>
    ggplot(aes(nmax)) +
    geom_histogram(bins = 20) +
    geom_segment(data = crazy_abund,
                 mapping = aes(x = n, y = rep(10, 3),
                               xend = n, yend = rep(0, 3)),
                 color = "red",
                 arrow = arrow(length = unit(1, "char")),
                 inherit.aes = FALSE) +
    xlab("Maximum abundances") +
    ylab("Count") +
    this_ms_look()
```

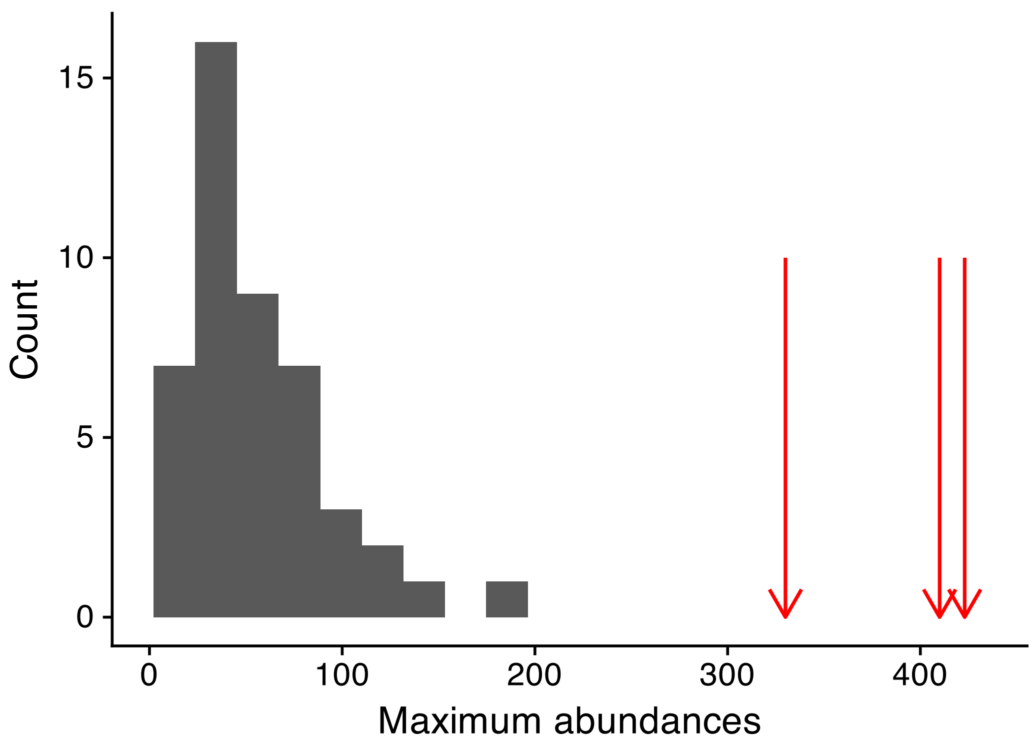

Figure 2

Because these samples are so unusual we decided to drop them from all further analyses presented in the main text. However, we retain the outlier samples in the supplement to validate that our results are not biased by their removal.

```
# remove from mete results
arth_mete_out <- arth_mete
arth_mete <- filter(arth_mete,
                    !((site %in% outliers$site) & (tree %in% outliers$tree)))

# remove from arth data frame
arth_out <- arth
arth <- filter(arth,
               !((site %in% outliers$site) & (tree %in% outliers$tree)))
```

Deviations from METE across the chronosequence

Now we analyze how deviations from METE behave across the chronosequence. The following code produces the analysis of a hump-shaped versus flat relationship between METE deviation and flow age.

```
# quadratic model of sad z^2 across age
sad_rma_chrono <- site_means_lm(log(z2_sad) ~
                                    log(flow_age_my) + I(log(flow_age_my)^2) +
                                    (1 | site),
                                data = arth_mete)

# quadratic model of ipd z^2 across age
ipd_rma_chrono <- site_means_lm(log(z2_ipd) ~
                                    log(flow_age_my) + I(log(flow_age_my)^2) +
                                    (1 | site),
                                data = arth_mete)
```

Likelihood ratio tests (LRT) can evaluate the significance of the quadratic model

```
# likelihood ratio tests for sad and ipd across chrono
sad_lrt_chrono <- anova(sad_rma_chrono)
sad_lrt_chrono
```

```
Test of Moderators (coefficients 2:3):
QM(df = 2) = 20.5604, p-val < .0001
```

```r
ipd_lrt_chrono <- anova(ipd_rma_chrono)
ipd_lrt_chrono
```

```
Test of Moderators (coefficients 2:3):
QM(df = 2) = 3.3629, p-val = 0.1861
```

The quadratic model for SAD z^2-values is supported by the LRT with a P-value < 0.0001, but the quadratic model for IPD z^2-values did not have a significant P-value. However, the 95% CI for the quadratic term in the IPD z^2-values model has the majority of its range below zero, indicating the data do tend toward a quadratic shape, just with greater variance compared to the quadratic model for SAD z^2-values.

```r
ipd_rma_chrono
```

```
Fixed-Effects with Moderators Model (k = 5)

I^2 (residual heterogeneity / unaccounted variability): 0.00%
H^2 (unaccounted variability / sampling variability):   0.22
R^2 (amount of heterogeneity accounted for):            77.20%

Test for Residual Heterogeneity:
QE(df = 2) = 0.4328, p-val = 0.8054

Test of Moderators (coefficients 2:3):
QM(df = 2) = 3.3629, p-val = 0.1861

Model Results:

                   estimate      se     zval    pval    ci.lb   ci.ub
intrcpt             -0.0977  0.4763  -0.2051  0.8375  -1.0312  0.8358
site_log_age        -0.4812  0.2694  -1.7865  0.0740  -1.0092  0.0467  .
I(site_log_age^2)   -0.0664  0.0366  -1.8164  0.0693  -0.1380  0.0052  .

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The following code produces the visualization reported in the main text of deviation from METE across flow ages.

```
# prepare data frame of predicted values for plotting
x <- range(arth$flow_age_my) |>
    log() |>
    (\(x) seq(x[1], x[2], length.out = 50))() |>
    exp() |>
    data.frame(flow_age_my = _)

sad_hat <- site_means_predict(sad_rma_chrono, newdata = x)
ipd_hat <- site_means_predict(ipd_rma_chrono, newdata = x)

z2_chrono_pred <- bind_rows(z2_sad = data.frame(x, sad_hat),
                            z2_ipd = data.frame(x, ipd_hat),
                            .id = "metric") |>
    select(!se) |>
    rename(z2 = pred,
           z2_lo = ci.lb,
           z2_hi = ci.ub) |>
    mutate(across(starts_with("z2"), exp))


# plotting
fig_z2_chrono <- pivot_longer(arth_mete, c(z2_sad, z2_ipd),
             names_to = "metric", values_to = "z2") |>
    ggplot(aes(flow_age_my, z2, color = site_name)) +
    geom_jitter(width = 0.1) +
    # confidence envelope
    geom_ribbon(data = z2_chrono_pred,
                mapping = aes(x = flow_age_my,
                              ymin = z2_lo,
                              ymax = z2_hi),
                color = "transparent", alpha = 0.4) +
    # central tendency
    geom_line(data = z2_chrono_pred,
              mapping = aes(flow_age_my, z2),
              color = "black") +
    # facet by IPD or SAD and trick strip label into axis label
    facet_grid(
        rows = vars(metric),
        labeller = as_labeller(
            c(z2_ipd = "\"IPD\"~z^2*\"-value\"",
              z2_sad = "\"SAD\"~z^2*\"-value\""),
            label_parsed
```

```
        ),
        switch = "y"
    ) +
    this_ms_look(log_x = TRUE, log_y = TRUE) +
    theme(axis.title.y = element_blank(),
          strip.background = element_blank(),
          strip.placement = "outside")

fig_z2_chrono
```

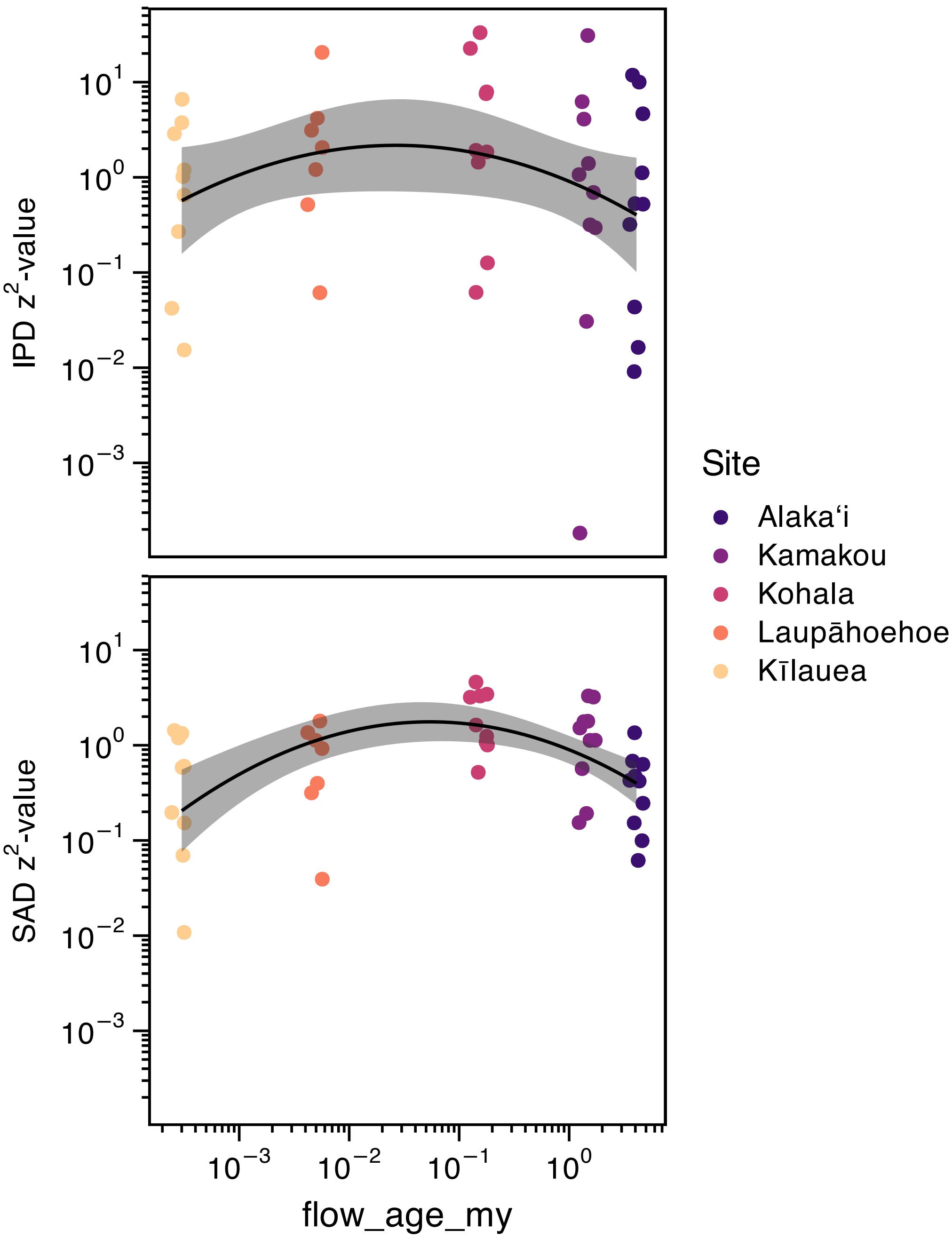

Figure 3: Deviations from METE as measured by z^2 across the chronosequence. Discussed in more detail in the main text.

How do observed patterns deviate from METE?

Next we determine how the observed data deviate from the METE predictions. To begin we look at two exemplar SADs and two exemplar IPDs representing low and high deviation from theory (Figure 4).

```r
# function to summarize theoretical and observed distributions
# given one sample `x`

mk_summ <- function(x) {
    esf <- meteESF(x$species_code, x$abundance, x$metabolic_rate)

    list(sad = sad(esf),
         ipd = ipd(esf)) |>
        lapply(function(x) {
            thr <- samp_mete(x, n = 1000) |>
                summarize_samp_mete()

            obs <- data.frame(rank = seq_along(x$data),
                              abund_obs = x$data)

            full_join(thr, obs)
        }) |>
        bind_rows(.id = "dist")
}



expl_samps <- data.frame(site = c("KA", "MO"),
                         tree = c(5, 9),
                         deviation = c("lo", "hi")) |>
    left_join(arth) |>
    (\(x) split(x, x$deviation, drop = TRUE))() |>
    lapply(mk_summ) |>
    bind_rows(.id = "deviation")
```

```
Joining with `by = join_by(site, tree)`
Joining with `by = join_by(rank)`
Joining with `by = join_by(rank)`
Joining with `by = join_by(rank)`
Joining with `by = join_by(rank)`
```

```r
fig_how_diff <- ggplot(expl_samps, aes(rank, m)) +
    geom_point(aes(rank, abund_obs)) +
    geom_ribbon(aes(ymin = lo, ymax = hi),
                fill = "#489BC5", alpha = 0.75) +
    geom_line(color = "#489BC5") +
    facet_grid2(
        rows = vars(dist), cols = vars(deviation),
        scales = "free", independent = "x",
        labeller = as_labeller(
            c(hi = "High deviation", lo = "Low deviation",
              ipd = "Metabolic rate", sad = "Abundance")
        ),
        switch = "y"
    ) +
    xlab("Rank") +
    ylab("") +
    this_ms_look(log_y = TRUE) +
    theme(strip.background.y = element_blank(),
          strip.placement.y = "outside",
          strip.text.y = element_text(size = 14))

fig_how_diff
```

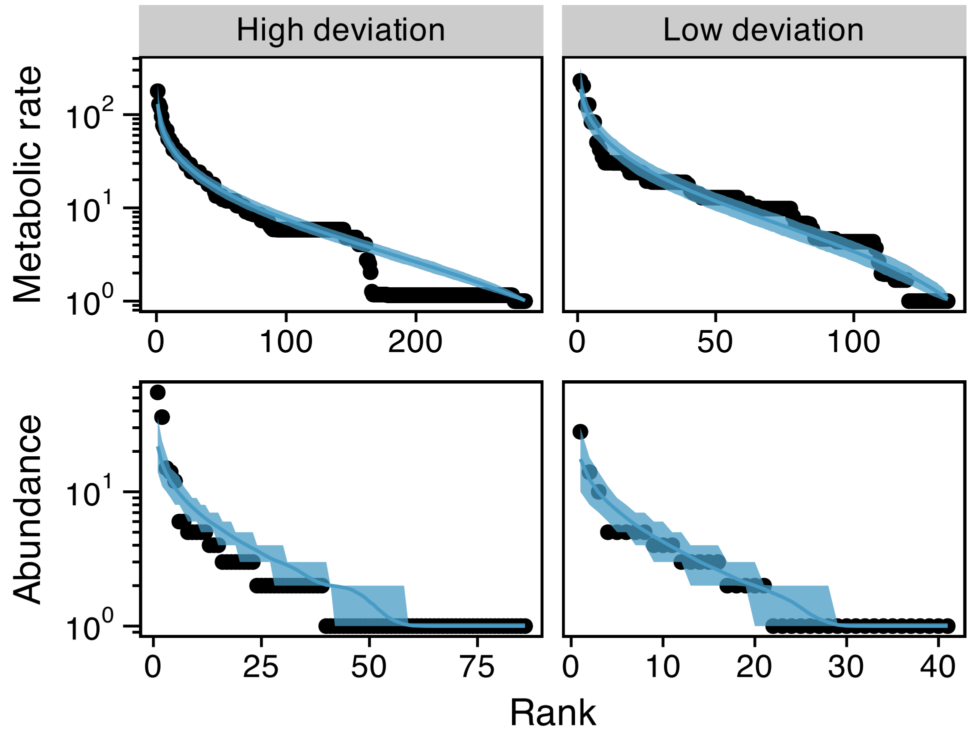

Figure 4: Two exemplar IPDs and SADs showing high and low deviation from METE. The low deviation distributions come from the same sample at the Alakaʻi site while the high deviation distributions come from the same sample at the Kamakou site. Samples were choosen to represent ranges of deviation while still being similar in state variables

Looking across all samples (Figure 5) we see that deviations between observed and theoretical SADs is consistently in the direction of under-predicted both rarity and dominance (i.e. observed SADs have more rarity and greater dominance than theoretical predictions). **?@fig-sad-how** shows that when METE predictions deviate from observed SADs, METE under-predicts both rarity and dominance compared to observed SADs. This means that observed SADs are more uneven than the METE prediction.

```
fig_how_z2 <- bind_rows(sad = select(arth_mete, site_name,
                                     z2 = z2_sad, rar_diff = n1_diff, dom_diff = nmax_diff),
          ipd = select(arth_mete, site_name,
                                     z2 = z2_ipd, rar_diff = p1_diff, dom_diff = pmax_diff),
          .id = "dist") |>
    pivot_longer(ends_with("diff"),
                 names_to = c("comp", ".value"),
                 names_sep = "_") |>
```

```r
    ggplot(aes(diff, z2, color = site_name)) +
    geom_point() +
    facet_grid2(rows = vars(dist), cols = vars(comp),
                scales = "free", independent = "x",
                labeller = as_labeller(
                    c(ipd = "\"IPD\"~z^2*\"-value\"",
                      sad = "\"SAD\"~z^2*\"-value\"",
                      dom = "\"Diff. in dominance\"",
                      rar = "\"Diff. in rarity\""),
                    label_parsed
                ),
                switch = "both") +
    xlab("") +
    ylab("") +
    this_ms_look(log_y = TRUE) +
    geom_vline(xintercept = 0, color = "gray") +
    theme(strip.background = element_blank(),
          strip.placement = "outside",
          strip.text.y = element_text(size = 14))


fig_how_z2
```

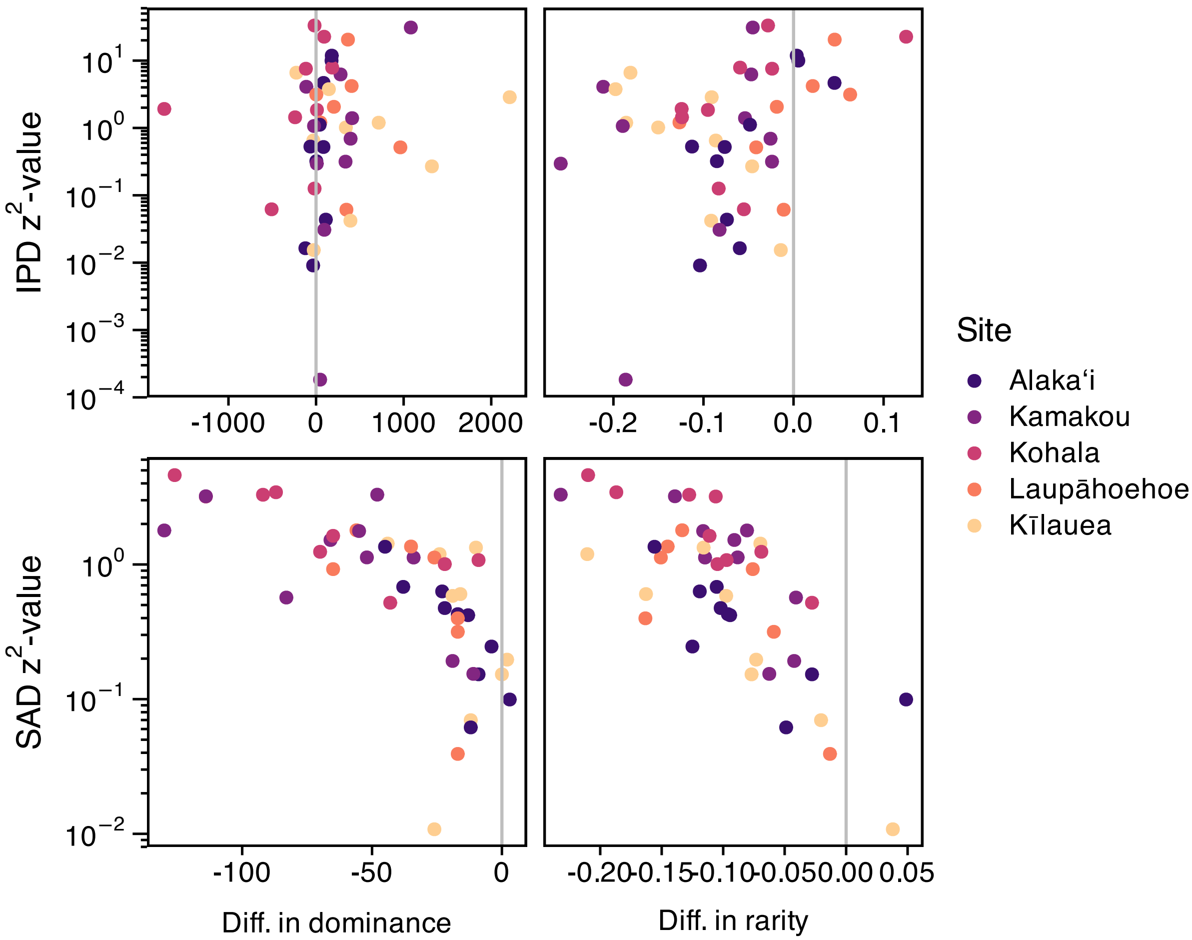

Figure 5: Relationship between z^2 measure of deviation from METE and different metrics describing how observed data and METE predictions diverge.

Comparing deviations from METE with potential explanatory variables

Deviation vs. proportion that are invasive

```
mete_by_non <- summarize(
    arth,
    prop_nspp_non_nat = n_distinct(
        species_code[!(origin %in% c("endemic", "indigneous"))]) /
        n_distinct(species_code),
    .by = site_name) |>
    full_join(arth_mete)
```

```
Joining with `by = join_by(site_name)`
```

```r
ggplot(mete_by_non, aes(prop_nspp_non_nat, z2_sad, color = site_name)) +
    geom_point() +
    this_ms_look(log_y = TRUE) +
    geom_hline(yintercept = exp(0.0474))

sad_rma_non <- site_means_lm(log(z2_sad) ~ prop_nspp_non_nat + (1 | site),
                             filter(mete_by_non, site != "VO"))

anova(sad_rma_non)
```

```
Test of Moderators (coefficient 2):
QM(df = 1) = 17.6582, p-val < .0001
```

```r
filter(mete_by_non, site == "VO") |>
    summarize(across(contains("nspp"), mean)) |>
    site_means_predict(sad_rma_non, newdata = _)
```

```
   pred     se   ci.lb  ci.ub
 0.3062 0.1878 -0.0619 0.6744
```

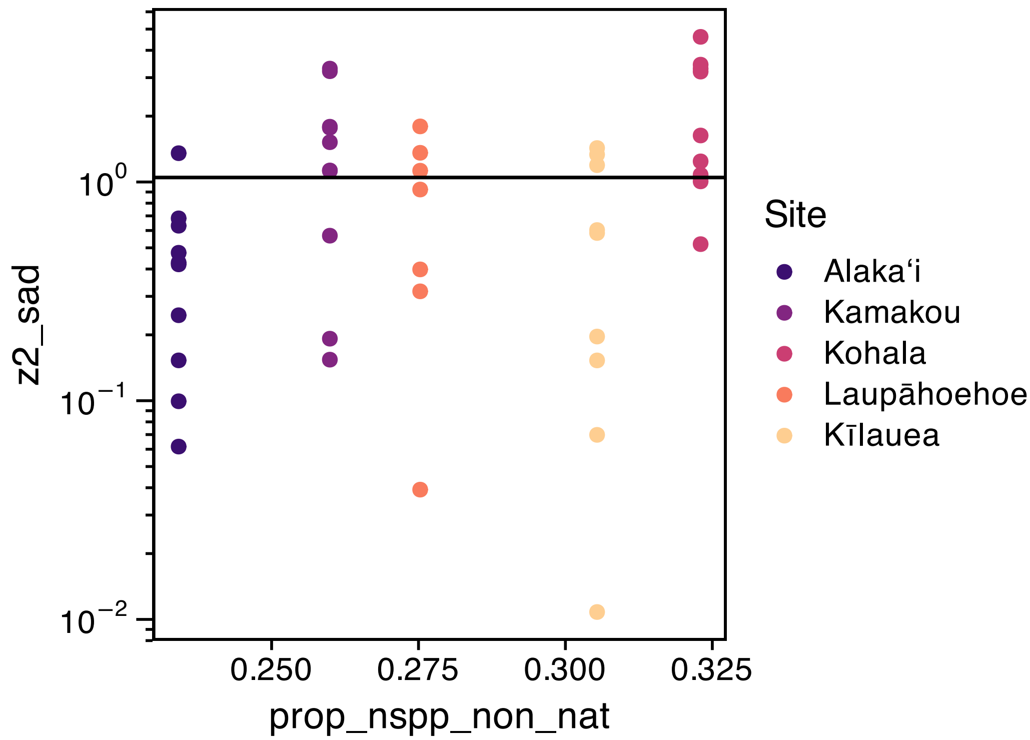

Figure 6

Deviation vs. dissimilarity

```
diss <- split(arth, arth$site) |>
    sapply(FUN = function(dat) {
        summarize(dat, abund = n(), .by = c(tree, species_code)) |>
            pivot_wider(names_from = species_code,
                        values_from = abund,
                        values_fill = 0) |>
            select(!tree) |>
            bray_diss_z()
    })


mete_by_diss <- mutate(arth_mete,
                       mean_diss = diss[site])
```

```r
ggplot(mete_by_diss, aes(mean_diss, z2_sad, color = site)) +
    geom_jitter(width = 0.25) +
    this_ms_look(log_y = TRUE)
ggplot(mete_by_diss, aes(mean_diss, z2_ipd, color = site)) +
    geom_jitter(width = 0.25) +
    this_ms_look(log_y = TRUE)


xy <- mete_by_diss |>
    summarize(z2_sad_mean = mean(log(z2_sad)),
              z2_sad_sd = sd(log(z2_sad)),
              n = n(),
              mean_diss = first(mean_diss),
              .by = site)




rma(yi = z2_sad_mean,
    vi = z2_sad_sd^2 / n,
    mods = ~ mean_diss,
    data = xy,
    method = "REML")
```

```r
site_bray_z <- split(arth, arth$site) |>
    sapply(function(dat) {
        comm <- summarize(dat,
                  abund = n(),
                  .by = c(tree, species_code)) |>
            pivot_wider(names_from = species_code,
                        values_from = abund,
                        values_fill = 0) |>
            select(!tree)
        g <- dat$origin[match(names(comm), dat$species_code)] %in%
            c("endemic", "indigenous")

        bray_part_z(as.matrix(comm), g)
    })
```